

Model Fitting of AMS-02 Cosmic-Ray Rigidity Spectra: the Leaky Box Model

Yuhan Chen

1 Introduction

Cosmic rays, consisting mainly of protons, makes up approximately 90% of all the cosmic rays that reach the surface of the earth. The remaining 10% are mainly helium nuclei. The researches on cosmic rays give us insight into both particle physics and high-energy astronomical objects. Considering the high velocity of the particles in cosmic rays, it is possible to study fundamental particles at such high energy level which is hard to replicated on the earth. It is available to study the properties of the high-density objects. The Alpha Magnetic Spectrometer(AMS) is a precision physics detector on the International Space Station [Aguilar M. et al. 2021]. The aim is to conduct long-term research on dark matter, antimatter and cosmic rays; it consists of a permanent magnet and an array of particle detectors to measure many physics properties including rigidity of the particles.

Based on the energies, cosmic rays are divided into two groups, namely low-energy cosmic rays and ultra-high energy cosmic rays(UHECRs). Even though the origin of the highest energy cosmic rays is still unknown, some possible sources of UHECRs includes accretion shocks in large-scale, AGN and neutron stars. [Kotera et al. 2011]. The dominant component of cosmic rays originates in the Galaxy [Kotera et al. 2011]. The main features are found in the spectrum of cosmic rays. The knee feature in the spectrum is caused by the escape of cosmic rays out of the Galaxy. The ankle feature at about 4EeV is possibly the result of the Galactic to extragalactic cosmic rays or the propagation loss assuming the cosmic rays are mainly consists of protons. Above 30Eev, the toe feature is caused by the interaction of the cosmic rays with the cosmic background radiation, known as Greisen-Zatsepin-Kuzmin(GZK) effect.

For research purpose of this report the cosmic rays are classified by primary cosmic rays and secondary cosmic rays. The primary cosmic rays are particles that come from outer space; they originates from cosmic rays that are produced by cosmic sources such as supernova, active galactic nuclei. The secondary cosmic rays are defined in two different ways in astrophysics, the first is defined by a decay of primary cosmic rays as they impact with the atmosphere. The interaction of cosmic rays with the atoms and molecules in the air produces lots of particles such as protons, neutrons, electrons, and photons(Bonomi et

al. 2020). However, in this report, the secondary cosmic rays are created when the primary cosmic rays passing through the interstellar medium. In these species, the helium, carbon, oxygen atoms are usually considered as primary cosmic rays, while Lithium, Berillium, Boron atoms are viewed as secondary species. The primary and secondary species have different flux spectrum since the mechanism they suffer are different. This is shown in our results in our project. Other species, such as nitrogen, contain both primary and secondary components. Therefore, in our model the parameter, gamma, for nitrogen are expected to lie between those parameters obtained for predominantly primary and predominantly secondary species.

2 Methods

2.1 Data sources

This report researches the rigidity spectra of Galactic cosmic-ray nuclei using the published AMS-02 flux measurements gathered by the Space Science Data Center (SSDC) cosmic-ray database. The objective is to estimate spectral shapes and information about the parameters of Broken power law in the Leaky-Box model. Leaky-box provides a framework for interpretation; the Diffusion is more appropriate at energy below 10^{17}eV . In Leaky-Box model it is approximate by leaky term [Strong et al. 2007].

The flux tables for protons, helium, lithium, beryllium, boron, carbon, nitrogen, oxygen, fluorine, silicon and iron are contained in Appendix. The proton, Li, Be, B, C, N and O tables originate from the AMS review [Aguilar et al. 2021c]; the He, Si, F and Fe tables are from separate publications [Aguilar et al. 2017; Aguilar et al. 2020; Aguilar et al. 2021a; Aguilar et al. 2021b]. The analysis includes eleven distinct species fits.

For each species, the rigidity R_i , differential flux y_i and its reported uncertainty σ_i from the AMS review are recorded. Rigidity is $R = \frac{pc}{|Z|e}$, where p is particle momentum, Ze its charge and c the speed of light. Fluxes are analyzed as differential fluxes per GV, with units $(\text{m}^2 \cdot \text{s} \cdot \text{sr} \cdot \text{GV})^{-1}$.

The solar wind and the magnetic field from the Sun have considerable effect on the propagation process of cosmic rays. The researches show that the solar modulation begins at about 30GV (Potgieter and Du T., 2013). A lower rigidity off-set around 45GV constrain the fits to a region where heliospheric modulation is within the measurement precision of AMS. The implemented thresholds are 50 GV for protons, 60 GV for carbon and 45 GV for the other nine fitted species. All points above corresponding thresholds are retained. Appendix A gives the actual sampled intervals and point counts.

The lower and upper bound of errors for flux in all eleven input tables are equal and positive; the common value is used as σ_i . The original data headers do not separately document the statistical/systematic components. The errors for rigidity are zero throughout the data adopted in this report. The original AMS data are gathered in different data bins. Zero horizontal errors do not

imply perfect rigidity precision.

2.2 The spectral model

Each species is fitted with the broken power law used in AMS spectral studies [Aguilar et al. 2015]:

$$\Phi(R; \boldsymbol{\theta}) = C \left(\frac{R}{R_p} \right)^\gamma \left[1 + \left(\frac{R}{R_0} \right)^{\Delta\gamma/s} \right]^s, \quad R_p = 45 \text{GV},$$

where $\boldsymbol{\theta} = (C, \gamma, \Delta\gamma, s, R_0)$. The parameter C sets the normalisation, R_0 is the characteristic transition rigidity, and s is related to the transition shape. R_p is a fixed reference scale, independent of the data-selection threshold.

All fitted lines have $\Delta\gamma > 0$ and $s > 0$. For these branches, the asymptotic slopes below and above the transition are γ and $\gamma + \Delta\gamma$, respectively. At bounded rigidity scope, direct differentiation gives the local slope

$$\gamma_{\text{eff}}(R) = \frac{d \ln \Phi}{d \ln R} = \gamma + \Delta\gamma \frac{q(R)}{1 + q(R)}, \quad q(R) = \left(\frac{R}{R_0} \right)^{\Delta\gamma/s}.$$

Thus the fitted γ is an asymptotic parameter rather than a slope uniformly over the observed range. The interpretations above are restricted to the positive- $\Delta\gamma$ solutions actually obtained.

2.3 Numerical estimation

In the reports, the measured fluxes are treated as independent Gaussian variables with fixed, heterogeneous uncertainties:

$$y_i = \Phi(R_i; \boldsymbol{\theta}) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_i^2).$$

In this report the fitted model is a working one for published flux estimates, rather than an event-count likelihood. In the following the fitting formulation explicitly clarifies the assumptions behind weighted fitting [Hogg, Bovy and Lang 2010]. Its likelihood function is

$$\ell(\boldsymbol{\theta}) = -\frac{1}{2} \sum_{i=1}^n \left[\frac{(y_i - \Phi(R_i; \boldsymbol{\theta}))^2}{\sigma_i^2} + \ln(2\pi\sigma_i^2) \right].$$

In the fitting process σ_i is treated as fixed, known value, maximising this likelihood is equivalent to minimising

$$\chi^2(\boldsymbol{\theta}) = \sum_{i=1}^n r_i(\boldsymbol{\theta})^2, \quad r_i(\boldsymbol{\theta}) = \frac{y_i - \Phi(R_i; \boldsymbol{\theta})}{\sigma_i}.$$

Though bin-to-bin systematic correlations are omitted, this method serves as an approximation and the data table leveraged in this report adjusted from

AMS-02 data release do not list the covariance matrix of errors. Such correlations can materially affect cosmic-ray parameter inference [Derome et al. 2019]; the degree of freedom test further illustrate the limitation to the methods.

All five parameters are varied simultaneously with the bounded algorithm in *scipy.optimize.least_squares* [Virtanen et al. 2020]. Initial values and parameter bounds are listed in Appendix A.2. Each species uses one starting point, with tolerance of the iteration $f_{tol}, x_{tol}, g_{tol}$ set to 10^{-13} . Optimiser termination condition is checked, but a multi-start search or global-optimum is not guaranteed.

2.4 Uncertainty estimation

Let $J_{ij} = \partial r_i / \partial \theta_j$ at the fitted points. The estimation parameters covariance from the Gauss–Newton curvature $J^T J$. For a full-rank, locally solution, the approximation is

$$\hat{V}_\theta \simeq (J^T J)^{-1}, \quad \text{SE}(\hat{\theta}_j) = \sqrt{(\hat{V}_\theta)_{jj}}.$$

The specific choices are recorded in Appendix A. Moreover, $J^T J$ is a Gauss approximation, not generally the exact observed Hessian of the nonlinear likelihood.

Parameter correlations and numerical condition numbers are used to estimate sensitivity, with the qualification that an unscaled condition number depends on parameter units. The solution for Li and N species reach the upper bound $s = 1$, Si reaches the lower bound $s = 0.003$, and F reaches $R_0 = 50$ GV. Symmetric local errors for these cases are provisional. Profile-likelihood analysis is available for weak parameter constraints [Raue et al. 2009], it is not implemented in this project.

2.5 Fitting diagnostics

The method is

$$\chi_{\text{red}}^2 = \frac{\chi^2(\hat{\theta})}{n - 5}.$$

Here $n - 5$ is the degrees of freedom. Its chi-square interpretation is only approximation for identifiable nonlinear fits with independent Gaussian errors. Correlated systematics, bounds problems are limitations in fitting inpretation. In particular, a very small reduced chi-square, which is detected after fitting or a large upper-tail p -value does not establish that the physical model is correct.

For presentation purpose, fluxes predictions and vertical uncertainties are multiplied by the same factor $R^{2.7}$. Parameter estimation is conducted on the unscaled data.

2.6 Physics Interpretation

In a simplified steady-state, high-rigidity limit, take a primary injection spectrum $Q_P(R) \propto R^{-\alpha}$ and an escape time $\tau_{\text{esc}}(R) \propto R^{-\delta}$. If energy losses, convection, reacceleration and destruction are negligible during the escape time period, then $N_P \propto Q_P \tau_{\text{esc}}$. If secondary cosmic rays are created dominantly by primary parents, another escape-time factor gives

$$N_P(R) \propto R^{-(\alpha+\delta)}, \quad N_S(R) \propto R^{-(\alpha+2\delta)}.$$

For an approximately isotropic relativistic population, $\Phi(R) \simeq \frac{vN(R)}{4\pi}$ and $v \simeq c$, the corresponding flux slopes are approximately the same. As alluded to above, these relations motivate primary species to compare with secondary ones, but do not specify a full description or identify injection and diffusion parameters separately [Strong et al. 2007; Genolini et al. 2019].

Accordingly, the present fittings provide a comparison of spectral shapes. They do not, by themselves, measure a universal diffusion index, or reject the leaky-box framework. Such claims require an explicit joint physical model test, comparable rigidity intervals, and appropriate treatment of uncertainty. Nitrogen is a mixed primary–secondary species, which further limits interpretation through a single asymptotic index [Aguilar et al. 2021c].

3 Results and Conclusion

The detailed results of the model curve fitting is in Appendix B. The analysis is based on eleven species, using 309 retained measurements in total. The lower selection thresholds are 50 GV for protons, 60 GV for carbon and 45 GV for the remaining species. Across the selected intervals, the fitted curves describe the measurements with small standardised residuals and nominal reduced chi-square values between 0.0459 and 0.6485.

In all of the graphs, the flux decreases with rigidity. Multiplication by $R^{2.7}$ makes the differences in spectral shape clear. The He, C, O and Si curves rise at the high-rigidity regime of the scaled plots; the curve proton, Li, B and N decrease at relatively low rigidity regime and show an upward turn afterwards. Be and F retain decreasing scaled curves over the curve ranges.

The solutions have $\Delta\gamma > 0$. However, the adopted parameterisation with $s > 0$ imposes non-negative logarithmic curvature; an increase in fitted local slope is consequently not an independent test for spectral hardening.

Fitting agreement does not imply that all five model parameters are reliably determined. Together with heterogeneous variance and omitted systematic correlations, the findings have limitation in quantitative comparisons between species.

The main finding in this report is that the spectral index parameter is a species-dependent feature. The scope of this project is a reproducible spectral description. Further work are required to compare species over a wide rigidity selection and elaborate statistical analysis of the AMS-02 original release.

4 References

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5 Appendix

A Data source and fitting settings

A.1 Data proven and selection

Table 1: Data sources and rigidity scope. The cut is the lower threshold; the retained interval contains the actual fitted point coordinates.

Species	Data publication	Observation interval (YYYY-MM-DD)	Cut (GV)	Retained R (GV)	n
p	[Aguilar et al. 2021c]	2011-05-19 to 2018-05-28	50	50.35–1465	30
He	[Aguilar et al. 2017]	2011-05-19 to 2016-05-26	45	46.8–2400	32
Li	[Aguilar et al. 2021c]	2011-05-19 to 2018-05-28	45	46.8–2300	31
Be	[Aguilar et al. 2021c]	2011-05-19 to 2018-05-28	45	46.8–2300	31
B	[Aguilar et al. 2021c]	2011-05-19 to 2018-05-28	45	46.8–2300	31
C	[Aguilar et al. 2021c]	2011-05-19 to 2018-05-28	60	62.55–2400	28
N	[Aguilar et al. 2021c]	2011-05-19 to 2018-05-28	45	46.8–2300	31

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Table 1 (continued)

Species	Data publication	Observation interval (YYYY-MM-DD)	Cut (GV)	Retained R (GV)	n
O	[Aguilar et al. 2021c]	2011-05-19 to 2018-05-28	45	46.8–2400	32
F	[Aguilar et al. 2021a]	2011-05-19 to 2019-11-01	45	48.65–2050	16
Si	[Aguilar et al. 2020]	2011-05-19 to 2018-05-26	45	46.8–2100	31
Fe	[Aguilar et al. 2021b]	2011-05-19 to 2019-11-01	45	48.65–2100	16

A.2 Parameter bounds and initial values

The common bounds are

$$\begin{aligned}
-4.5 &\leq \gamma \leq -1.5, \\
-0.5 &\leq \Delta\gamma \leq 1.5, \\
0.003 &\leq s \leq 1.0, \\
50 &\leq \frac{R_0}{\text{GV}} \leq 3000.
\end{aligned} \tag{1}$$

For carbon, the upper bound on $\Delta\gamma$ is 1.0 instead of 1.5. Bounds on the flux normalisation C and the initial parameter values are listed in Table 2.

Table 2: Normalisation bounds, initial values and covariance calculation. The superscript “init” denotes an optimiser starting value.

Species	C bounds	C^{init}	γ^{init}	$\Delta\gamma^{\text{init}}$	s^{init}	R_0^{init} (GV)
p	$[10^{-4}, 5]$	0.415	−2.79	0.13	0.024	336
He	$[10^{-6}, 1]$	0.09	−2.75	0.17	0.05	350
Li	$[10^{-9}, 0.1]$	3.5×10^{-4}	−3.1	0.30	0.05	300
Be	$[10^{-8}, 0.1]$	2.0×10^{-4}	−3.0	0.30	0.05	300
B	$[10^{-8}, 0.1]$	5.05×10^{-4}	−3.04	0.30	0.05	300
C	$[10^{-6}, 1]$	3.1×10^{-3}	−2.756	0.17	0.05	340
N	$[10^{-9}, 0.1]$	5.5×10^{-4}	−3.0	0.30	0.05	300
O	$[10^{-8}, 1]$	3.0×10^{-3}	−2.7	0.15	0.05	300
F	$[10^{-9}, 0.01]$	3.5×10^{-5}	−3.2	0.40	0.10	300
Si	$[10^{-9}, 0.1]$	5.0×10^{-4}	−2.7	0.15	0.05	300

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Table 2 (continued)

Species	C bounds	C^{init}	γ^{init}	$\Delta\gamma^{\text{init}}$	s^{init}	R_0^{init} (GV)
Fe	$[10^{-8}, 0.1]$	4.0×10^{-4}	-2.7	0.15	0.05	300

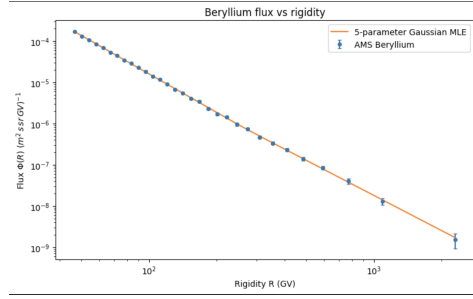
A.3 Numerical analysis of the fittings

Table 3: Numerical number of the eleven independent fits. The residual degrees of freedom are $\nu = n - 5$, and $\chi_{\text{red}}^2 = \chi^2/\nu$. An active bound indicates that a fitted parameter reaches an imposed limit.

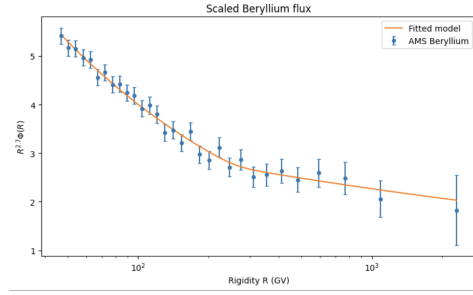
Species	χ^2	Nominal ν	χ_{red}^2	Active bound
p	1.3828	25	0.0553	None
He	2.0422	27	0.0756	None
Li	11.8567	26	0.4560	$s = 1.0$ (upper)
Be	12.0139	26	0.4621	None
B	4.6053	26	0.1771	None
C	4.0510	23	0.1761	None
N	8.6973	26	0.3345	$s = 1.0$ (upper)
O	1.2398	27	0.0459	None
F	7.1333	11	0.6485	$R_0 = 50$ GV (lower)
Si	6.8845	26	0.2648	$s = 0.003$ (lower)
Fe	3.5272	11	0.3207	$s = 0.003$ (lower)

B Spectral Fits and Residual Diagnostics

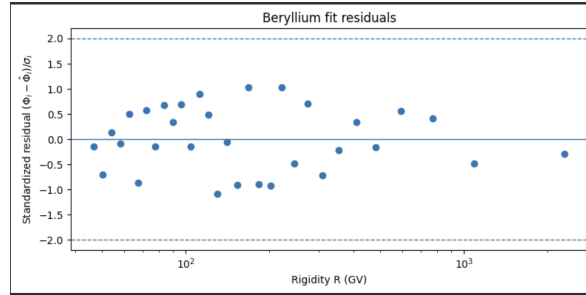
The following figures present the fitted rigidity spectra, scaled flux representations, and residual diagnostics for all cosmic-ray species analysed in this work. These plots provide supplementary information for assessing the quality of the spectral fits and the systematic differences between primary, secondary, and mixed cosmic-ray species.



(a) Measured flux as a function of rigidity

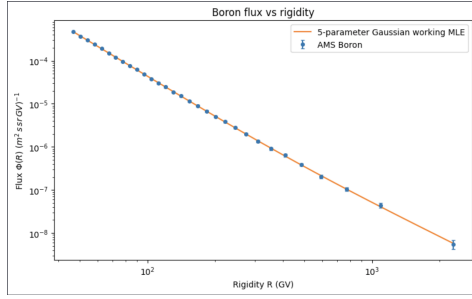


(b) Scaled flux representation.

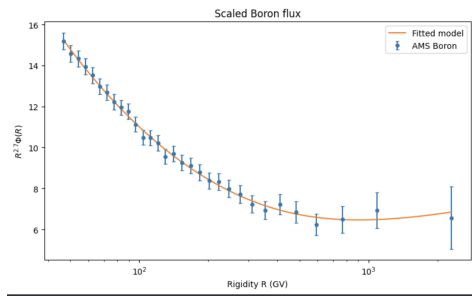


(c) Standardised residuals of the fitted model.

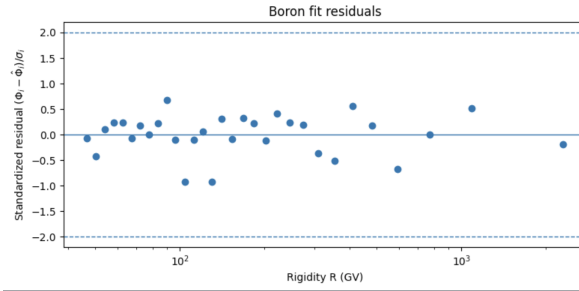
Figure 1: Spectral fitting results for Beryllium.



(a) Measured flux as a function of rigidity

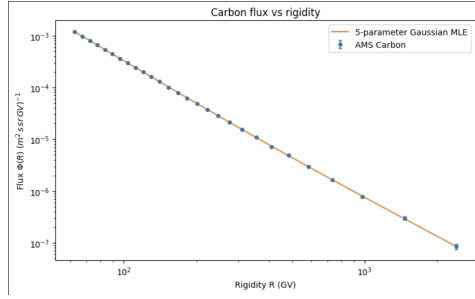


(b) Scaled flux representation.

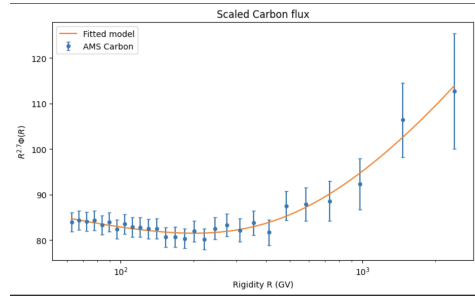


(c) Standardised residuals of the fitted model.

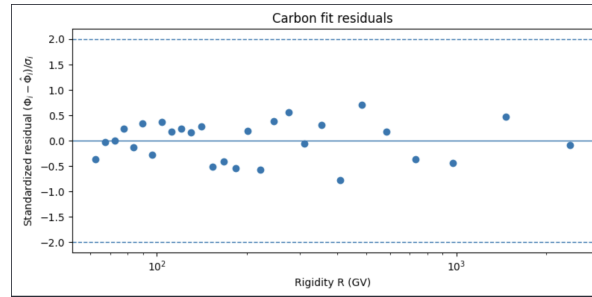
Figure 2: Spectral fitting results for Boron.



(a) Measured flux as a function of rigidity

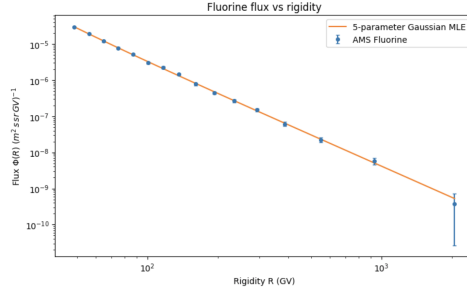


(b) Scaled flux representation.

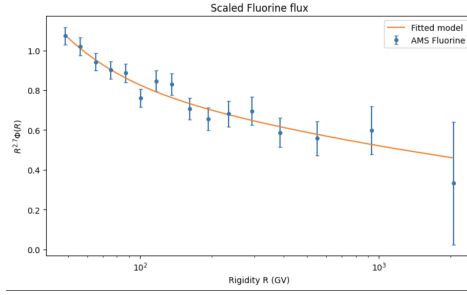


(c) Standardised residuals of the fitted model.

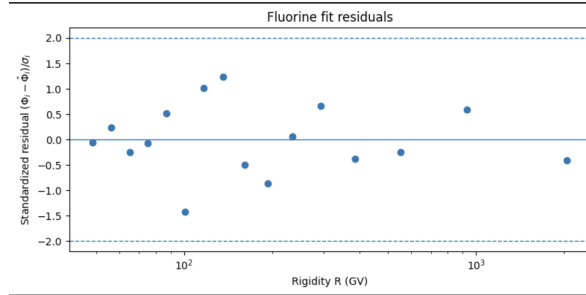
Figure 3: Spectral fitting results for Carbon.



(a) Measured flux as a function of rigidity

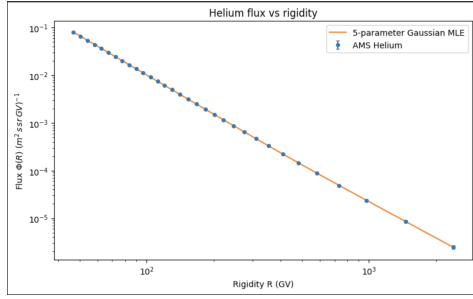


(b) Scaled flux representation.

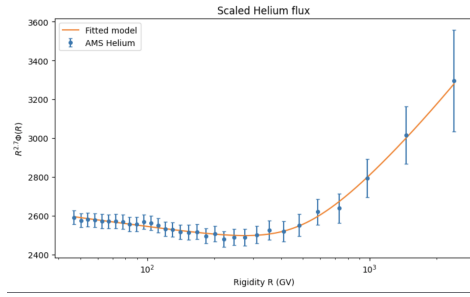


(c) Standardised residuals of the fitted model.

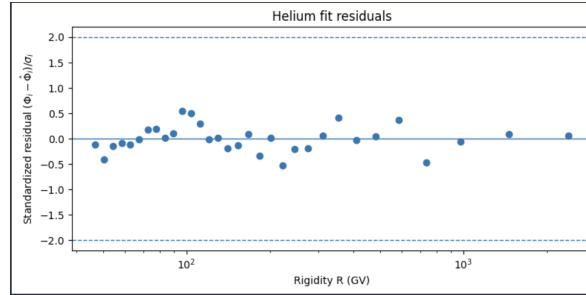
Figure 4: Spectral fitting results for Fluorine.



(a) Measured flux as a function of rigidity

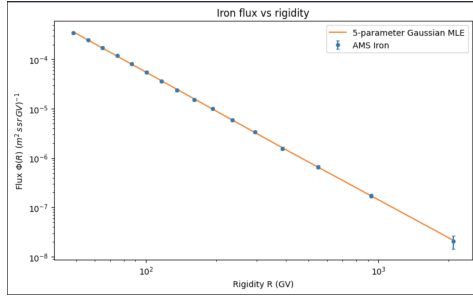


(b) Scaled flux representation.

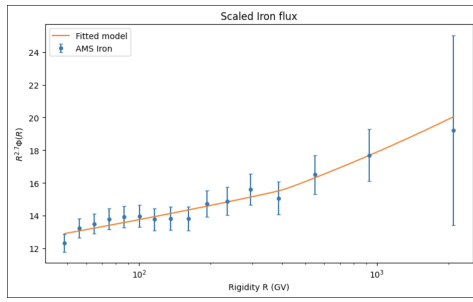


(c) Standardised residuals of the fitted model.

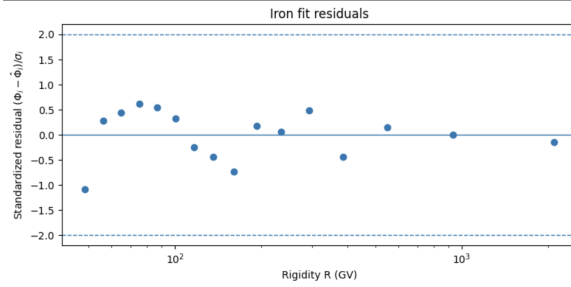
Figure 5: Spectral fitting results for Helium.



(a) Measured flux as a function of rigidity

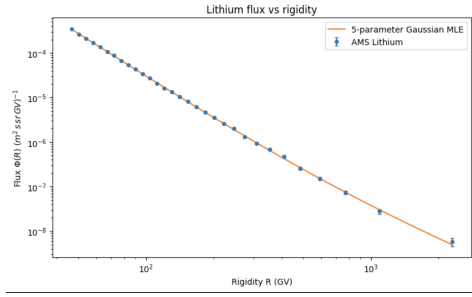


(b) Scaled flux representation.

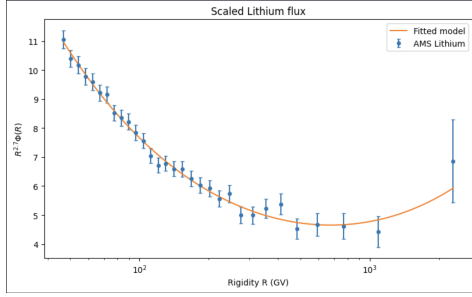


(c) Standardised residuals of the fitted model.

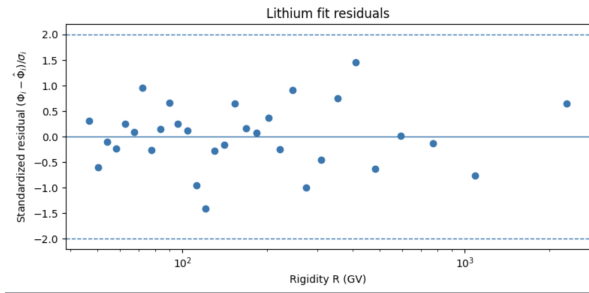
Figure 6: Spectral fitting results for Iron.



(a) Measured flux as a function of rigidity

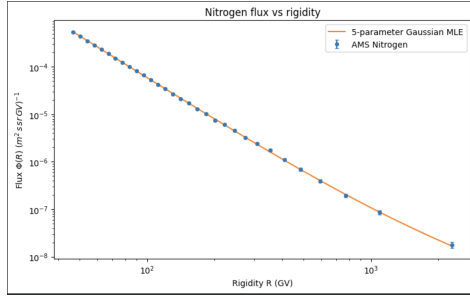


(b) Scaled flux representation.

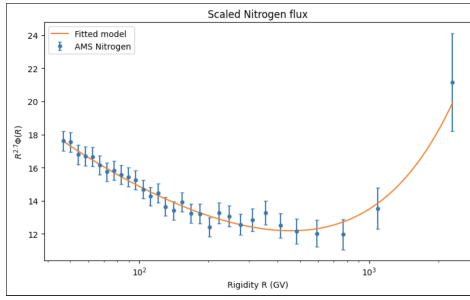


(c) Standardised residuals of the fitted model.

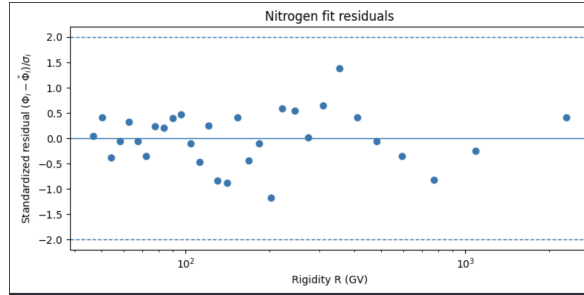
Figure 7: Spectral fitting results for Lithium.



(a) Measured flux as a function of rigidity

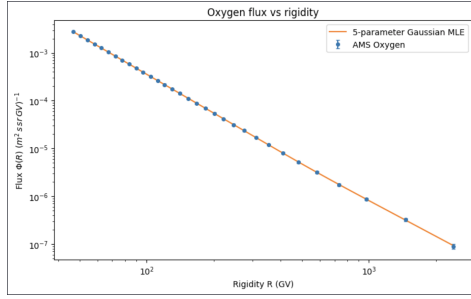


(b) Scaled flux representation.

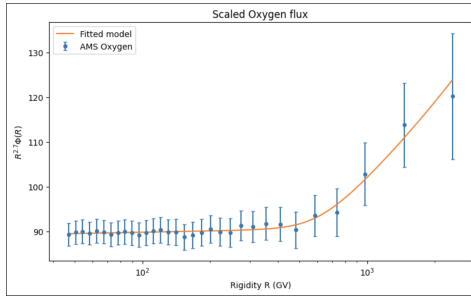


(c) Standardised residuals of the fitted model.

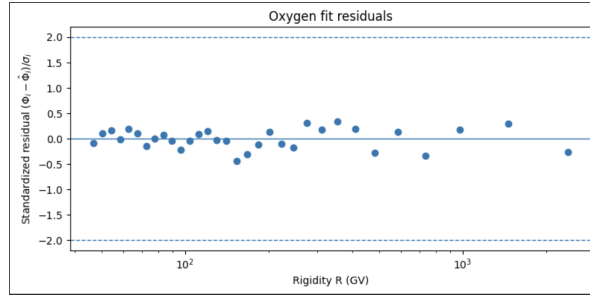
Figure 8: Spectral fitting results for Nitrogen.



(a) Measured flux as a function of rigidity

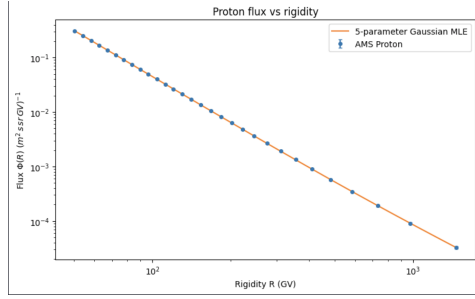


(b) Scaled flux representation.

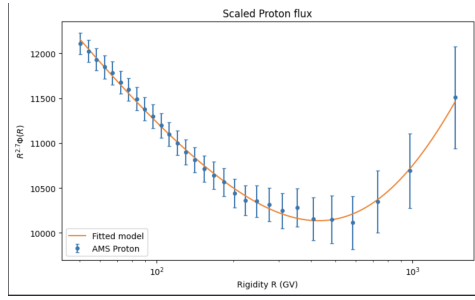


(c) Standardised residuals of the fitted model.

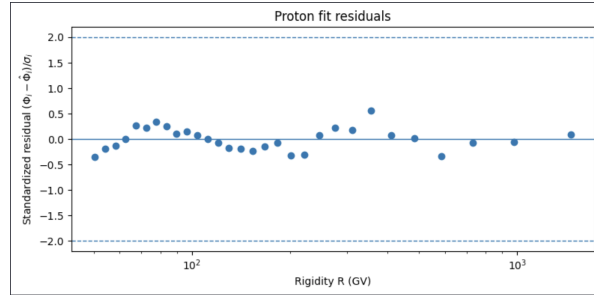
Figure 9: Spectral fitting results for Oxygen.



(a) Measured flux as a function of rigidity

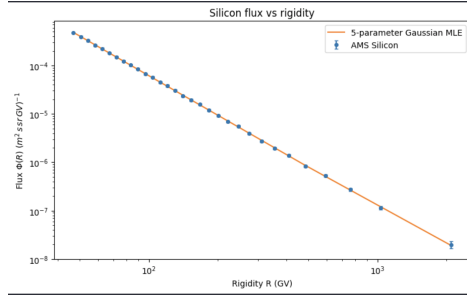


(b) Scaled flux representation.

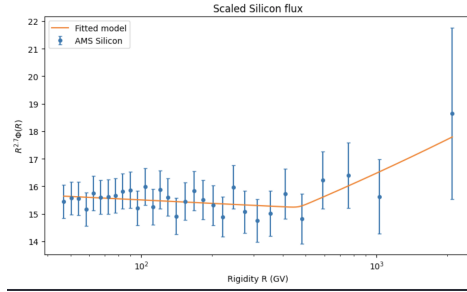


(c) Standardised residuals of the fitted model.

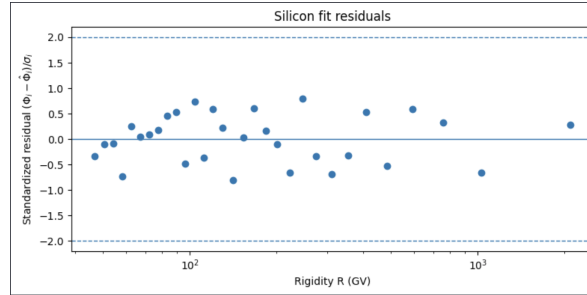
Figure 10: Spectral fitting results for Proton.



(a) Measured flux as a function of rigidity



(b) Scaled flux representation.



(c) Standardised residuals of the fitted model.

Figure 11: Spectral fitting results for Silicon.